

Positive-Case Count and Synthetic Data Fidelity: A CART-Based Analysis of MIMIC-IV Clinical Data

William Chen
University of California, Irvine
Instructor: Prof. Sam Behseta
DATA298P

Abstract

A logistic fit’s precision is set by the number of positive cases rather than by the sample size. This report asks whether the same threshold governs a generator, using CART-based synthesis (**synthpop**) on MIMIC-IV ($n = 70,954$ ICU stays) and two sweeps that hold the generator fixed and vary only the input: one raises the rare stratum’s positive count k at fixed sample size, the other raises the sample size n at fixed k , which together separate count from prevalence. It does, but the generator adds no threshold of its own. Over increases of comparable size, raising k drives divergence to zero between counts of forty and eighty and cuts interquartile width by 42–66% in all six series, while raising n moves neither. Two of the four quantities never involve the synthesizer at all, so the threshold belongs to the stratum and **synthpop** inherits it rather than imposing a second one. What synthesis contributes instead is a bias no count removes: in a companion simulation with a known data-generating process, the synthetic median recovers 99% of a true interaction of 0.5 but only 41% of one of 3, at a count where neither arm fails. The finding transfers as a mechanism and not as a remedy. The interactions available in MIMIC-IV are small enough that a 500-row subsample carries 6–7% power to detect them, which is what the sweeps returned: the smallest interaction such a subsample could detect is seven to twelve times the one MIMIC-IV holds, and eighty percent power would need 23,384 to 67,238 rows. No count reachable inside a subsample of this size makes the estimate usable. Two recommendations follow: establish that a target effect is estimable in the source data before attributing any discrepancy to the generator, and calibrate variable-selection diagnostics against a permutation null, whose own rate here moves from 0.00 to 0.32 with class balance alone.

1 Introduction

Synthetic data is often proposed as a way to share clinical records without exposing patients, but whether it preserves the relationships researchers care about is a separate question from whether it looks realistic overall. A prior analysis of the **birthwt** dataset found that under Bayesian logistic regression, posterior stability depends on the number of positive cases for a predictor rather than on overall sample size. That a logistic fit’s precision is set by the event count is itself established [1, 2], and the conditions under which small event counts produce separation have been characterised since [3]. What is not established is whether the same threshold governs a *generator*. Whether a CART-based synthesizer preserves a rare-stratum interaction is a question about which variables a tree spends a split on, not about the curvature of a likelihood, and the two need not share a threshold.

This report asks whether the same mechanism holds for CART-based sequential synthesis. The generator is held fixed throughout; the question is not which tree-based method performs best but how one of them behaves when the stratum it must reproduce is small. Because positive count and prevalence move together whenever sample size is fixed, a single sweep cannot attribute a change to either, so two complementary sweeps are run. Code is available at [GitHub Repository](#).

2 Data

2.1 Cohort

MIMIC-IV v2.2 [8], restricted to hospital admissions with an associated ICU stay. One row per ICU stay (73,181 stays) was built from `icustays`, `patients`, `admissions`, and `diagnoses_icd`, the last supplying the comorbidity flags. Dropping the 3% of stays with no first-day creatinine leaves 70,954 rows; all subsampling draws from this pool. Those 2,227 stays are not missing at random. They die in hospital at 28.1% against 10.9% for the stays kept, their median length of stay is 0.73 days against 1.96, and they carry every comorbidity less often: CHF 15.4% against 27.7%, Diabetes 21.7% against 30.5%, COPD 7.9% against 11.3%. The pattern is the one a first-day lab implies. A stay that ends within hours, by death or by discharge, has neither the time for the draw nor the encounter for the coding, so the lower comorbidity rates record thinner documentation rather than healthier patients. The pool is conditioned on surviving long enough to be measured, and $\hat{\beta}_{\text{full}}$ is accordingly the interaction among stays that clear that bar rather than among ICU stays at large. Both arms of every sweep are drawn from the same pool, so the comparison this report rests on is unaffected; what the selection limits is the clinical reading of the reference. Rows are not independent: the outcome and the comorbidity flags are recorded per admission, and the 70,954 stays span 64,308 admissions and 49,563 patients, so 30% of rows repeat a patient. Standard errors on the full-pool fit are clustered on patient. Age is MIMIC-IV’s `anchor_age`, which is the age at the anchor year rather than at admission and is censored above 89.

2.2 Comorbidity Screening

Five comorbidities were defined by simplified ICD-9/ICD-10 prefix groupings rather than a validated index such as Elixhauser or Charlson, since what the sweeps need is a binary flag with a known positive count rather than a calibrated severity score. They are counted by admission rather than by ICU stay, the unit the pool is indexed on: COPD 7,244, CHF 17,412, Diabetes 19,806, Renal Failure 25,369 and Hypertension 34,979. The three least common were carried forward. All five are common in the pool; the rare stratum is created by the subsampling below, not found in the data.

2.3 Continuous Variable Selection

Creatinine was chosen from five first-day lab candidates for its 97% coverage and its link to CHF through cardiorenal syndrome. The maximum in the first 24 hours was taken, capped at the 99.5th percentile (11 mg/dL) and left untransformed, so every creatinine coefficient below is per mg/dL.

3 Methods

3.1 Experimental Design

Both sweeps draw stratified subsamples, synthesize each with `synthpop::syn()` [7], and fit

$$\text{hospital_expire_flag} \sim \text{age} + \text{creatinine} + \text{comorbidity} + \text{comorbidity}:\text{creatinine}$$

to both the real subsample and the synthetic table, giving paired estimates $\hat{\beta}_{\text{real}}$ and $\hat{\beta}_{\text{syn}}$ of the interaction, at $M = 100$ Monte Carlo replicates per grid point.

Experiment	Held fixed	Varied	What moves
1. Count sweep	$n = 500$	$k \in \{5, 10, 20, 40, 80, 150, 250\}$	count and prevalence together
2. Size sweep	$k = 20$	$n \in \{250, 500, 1000, 2000, 4000\}$	prevalence only, 8% to 0.5%

3.2 Analyses

We conduct three analyses. (1) A full-pool fit of the interaction supplies the reference every subsample estimate is read against. (2) The two sweeps compare $\hat{\beta}_{\text{real}}$ against $\hat{\beta}_{\text{syn}}$ at each grid point on the median and interquartile width of the converged estimates, with the divergence and inestimable rates kept separate; Section 5 defines the two ways a fit fails. (3) A tree diagnosis refits the subsamples both sweeps draw with `rpart` under `synthpop::syn.cart`’s control parameters and records whether the comorbidity earns a split, against a likelihood-ratio test on the same rows that drops the comorbidity and its interaction together, so both are asked the same question. Every configuration is re-run with the comorbidity column shuffled, which leaves the positive count intact and destroys its association with the outcome, supplying the null. The two columns are compared across the whole grid by a conditional logistic model, `used ~ observed + observed:log2k_c + strata(pair)`, whose strata are the (comorbidity, replicate) pairs holding one observed and one shuffled outcome each, with $\log_2 k$ centred at $k = 40$ so that the main effect is read there; the fit runs over 4,200 rows and 585 splits.

Throughout, the synthetic arm is fitted the way an analyst who has been handed one synthetic file would fit it: a single synthetic copy, an ordinary GLM, and no correction for the variance synthesis itself contributes. Proper inference from synthetic data uses several copies and a combining rule [6], which would widen the synthetic intervals reported here. The naive fit is the one measured because it is the one a data recipient performs, but the comparison against the real arm is therefore between procedures as used, not between estimators of matched validity.

4 The Full-Pool Reference

MIMIC-IV carries no known ground truth, so the interaction fit on the whole pool stands in as the reference, $\hat{\beta}_{\text{full}}$, and every subsample estimate below is read against it. A companion simulation with a known data-generating process, where bias against the truth is measurable directly, is reported alongside the sweeps in Section 6.3. The three interactions differ in sign, CHF and Diabetes dampening creatinine’s main effect while COPD steepens it, and all three are roughly three to four times smaller than that main effect.

Comorbidity	Creatinine	$\hat{\beta}_{\text{full}}$	SE	p
CHF	0.208	−0.0497	0.0141	4.4×10^{-4}
Diabetes	0.234	−0.0709	0.0145	1.1×10^{-6}
COPD	0.188	+0.0668	0.0232	4.0×10^{-3}

Table 1: Comorbidity–creatinine interaction on the full pool, with the creatinine slope it modulates. Standard errors are clustered on patient.

5 Estimation Failure

5.1 Two Failure Modes

Fits at the smallest counts failed often enough to matter, and the failures turned out to be of two distinct kinds. We report their rates as a result rather than a filtering step: whether an estimate can be produced at all separates the two levers more sharply than the spread of those that succeed. Every quantile that follows is computed on converged fits only.

A fit is **inestimable** when the table holds no positives and the interaction term comes back as **NA**; **divergent** when the term is estimated but runs past $|2|$, marking a likelihood maximized at the boundary [5] rather than a real effect. Only divergence appears in the real arm, which is built to hold exactly k positives; the synthetic table generates its own column and can produce none.

The cutoff follows the scale of the target, around 0.05 in the full pool, and recomputing at 5 and 10 leaves every comparison pointing the same way. It under-detects at the smallest counts: at $k = 5$ between 64%

and 76% of real fits are separated, which a penalized fit [4] confirms, while [2] flags 18–34% of them, since `glm` stops after twenty-five iterations rather than running to the boundary. From $k = 20$ up the two agree. Both modes are reported as shares of all M replicates, so converged, inestimable and divergent sum to one.

5.2 Failure Under Each Lever

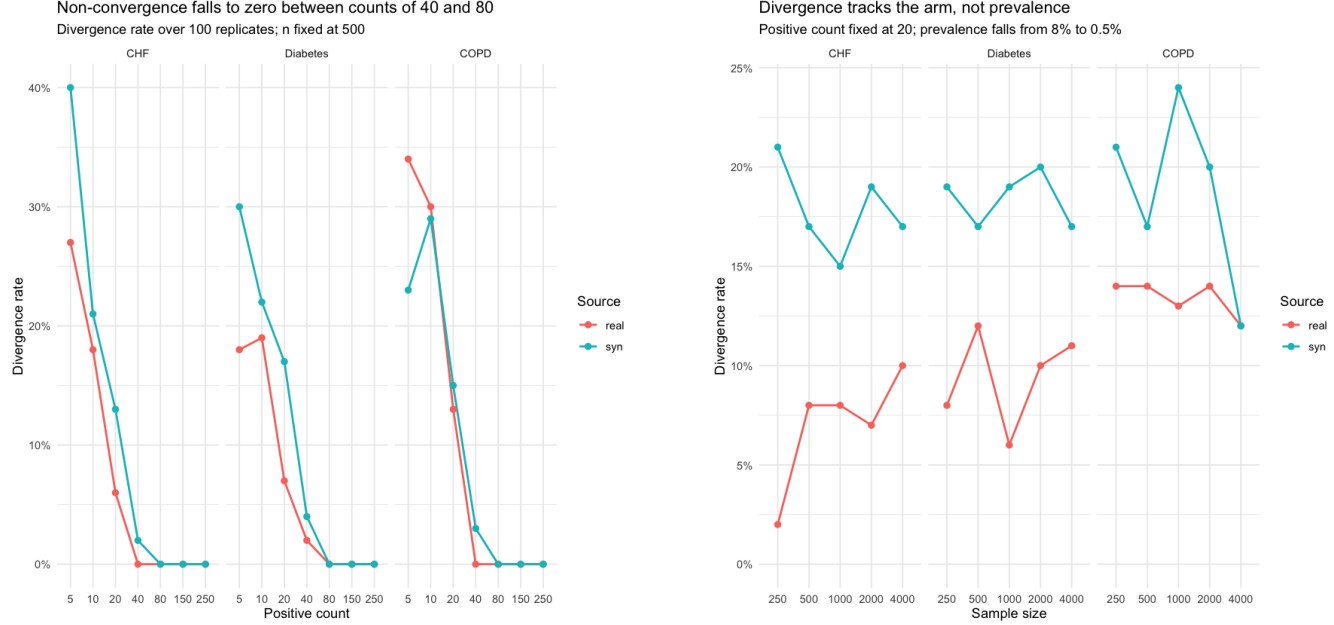


Figure 1: Divergence rate under each lever, three comorbidities $\times$ two sources in both panels. Left: positive count varying at $n = 500$. Right: sample size varying at $k = 20$, where prevalence falls from 8% to 0.5%. Inestimable fits occur on the synthetic side only and are reported in the text. Both panels use the [2] cutoff, which under-detects at $k \leq 10$ (Section 5), so the left panel’s first two counts are floors rather than rates.

Count sweep. Every series is at zero by $k = 80$, but not at the same count: the CHF and COPD real arms are already there at $k = 40$, where the other four sit at 2–4%, two to four replicates out of a hundred. Each zero is 0 of 100, which bounds a rate at 3.6% rather than establishing it as nil, so what the data supports is a threshold somewhere in $(40, 80]$ for each series rather than one shared value. Wherever it sits, it moves with neither prevalence, which varies nearly threefold across the three comorbidities, nor the sign of the interaction. Below it, divergence runs at 18–40% at $k = 5$ and is higher on the synthetic side in ten of the twelve failing cells. Both exceptions are COPD, at $k = 5$ and $k = 10$, where the real arm is the one that fails more often; Section 6.3 reproduces that reversal under a known truth and gives it a mechanism. The inestimable rate is narrower still: synthetic only, $k = 5$ only, at 4–6%.

Size sweep. No sample size does the same. Over the matched 8-fold increase in n every series changes by between -29% and $+25\%$, and none comes near zero (Figure 1, right). COPD is the plainest case: its real arm fails at 12–14% at every one of the five sample sizes. What does separate cleanly under this lever is the arm rather than the size: the synthetic rate is the higher of the two in fourteen of the fifteen cells and tied in the fifteenth, by 3 to 19 points, and no sample size closes the gap. The inestimable rate is zero in all fifteen cells, including at $n = 2000$, where prevalence matches the $k = 5$ cells that lost the stratum 4–6% of the time. Failure follows the count, and only the count.

6 Results

6.1 Experiment 1: Count Sweep

Spread narrows steadily with the count: from $k = 20$ to $k = 250$ every one of the six series loses 55% to 69% of its width, and none has flattened by the end of the range. It is still not enough. Even the narrowest cell, the CHF real arm at $k = 250$, has an interquartile width of 0.194 against its reference of 0.0497 (Table 1), nearly four times the quantity being estimated. The low end of the span rests on fewer replicates, 6% to 44% being discarded below $k = 40$, but all of them for diverging, so the decline is if anything understated.

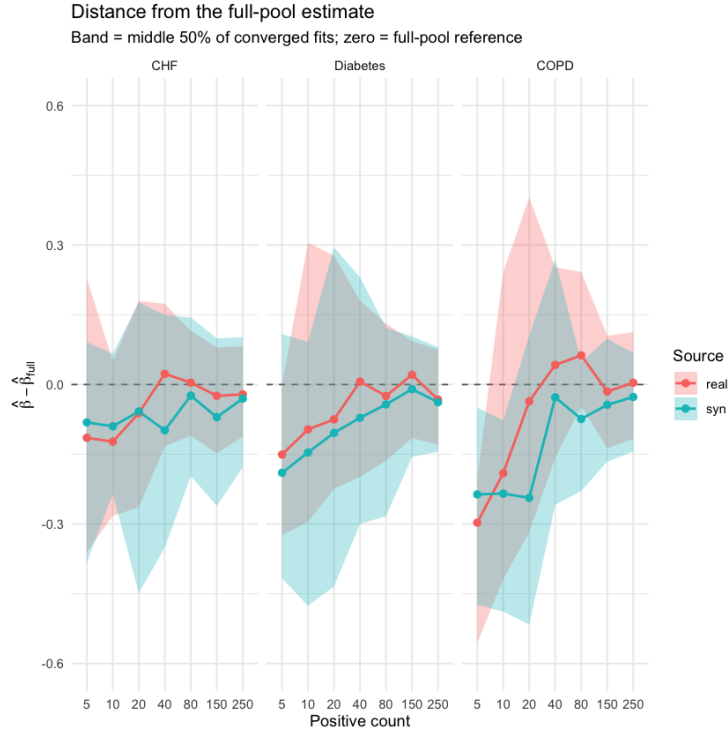


Figure 2: Comorbidity $\times$ creatinine, real vs. synthetic, plotted as distance from $\hat{\beta}_{\text{full}}$ so that zero is the full-pool value in all three panels. Band = middle 50% of converged fits.

What low counts cost is precision, not location: in 39 of the 42 cells the middle half of the estimates still contains $\hat{\beta}_{\text{full}}$, the three exceptions all COPD at $k \leq 10$. The two arms differ on both counts, and in the same direction. The synthetic median sits below the real one in 17 of the 21 counts, and the synthetic spread is the wider of the two.

How much wider is worth stating against a rate rather than against the real arm. The variance of an interaction draws on both groups, so it scales as $1/k + 1/(n - k)$, and holding n at 500 means raising k also thins the negatives. Rescaling the $k = 250$ width by that factor reproduces the real arm closely: the ratio of observed to predicted has a median of 0.97 and stays within 0.85 and 1.07 across the nine cells at $k \in \{40, 80, 150\}$; the $k = 250$ anchor is fixed at 1 by construction, and counts below 40 are excluded because divergence discards too many replicates there for a width to mean anything. The synthetic arm sits higher, median 1.14 against 0.97 and maximum 1.35, and the comparison that carries the weight is the paired one rather than the count of cells above 1: at every one of the nine matched cells, same comorbidity and same count, the synthetic ratio exceeds the real one (Figure 3; sign test $p = 0.004$). The count accounts for the real spread. The synthetic arm carries a further excess it does not explain, at every count where both arms can be read.

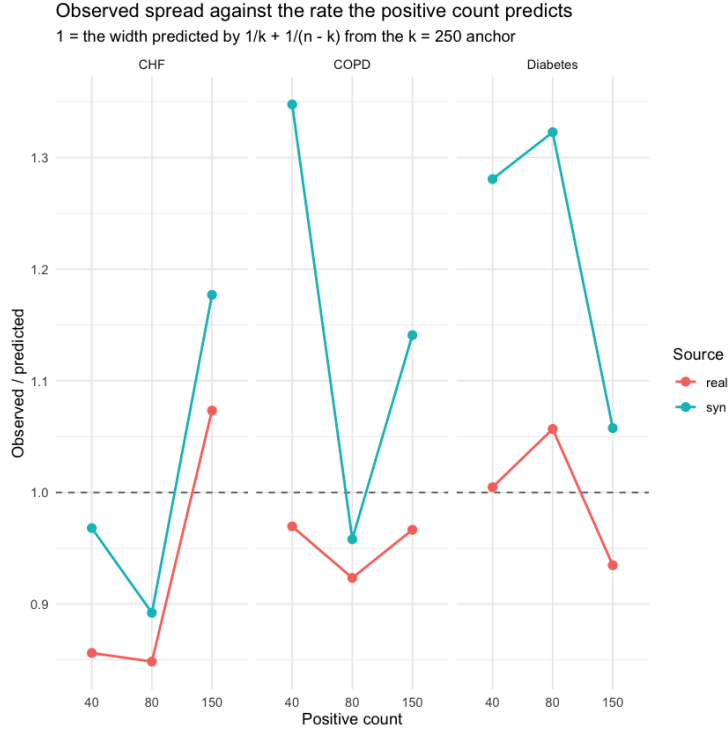


Figure 3: Interquartile width against the width the $1/k + 1/(n - k)$ rate predicts from the $k = 250$ anchor, which is fixed at 1 by construction and omitted. The real arm tracks the rate; the synthetic arm sits above it in six of the nine cells, with ranges that overlap.

6.2 Experiment 2: Size Sweep

The narrowing Experiment 1 found does not reappear under the other lever. Over roughly eightfold increases in each, k from 20 to 150 and n from 500 to 4000, the count cuts width by 42% to 66% in all six series while the sample size changes it by between -19% and $+10\%$ with no consistent sign (Figure 4). Across the full sixteen-fold range of n the widths never leave the band Experiment 1 recorded at $k = 20$. The count was held at 20, the last configuration in which it still binds: failure there runs 6–17% of replicates against 0–4% at $k = 40$ and none from $k = 80$ up. If sample size can substitute for count, this is where it would show.

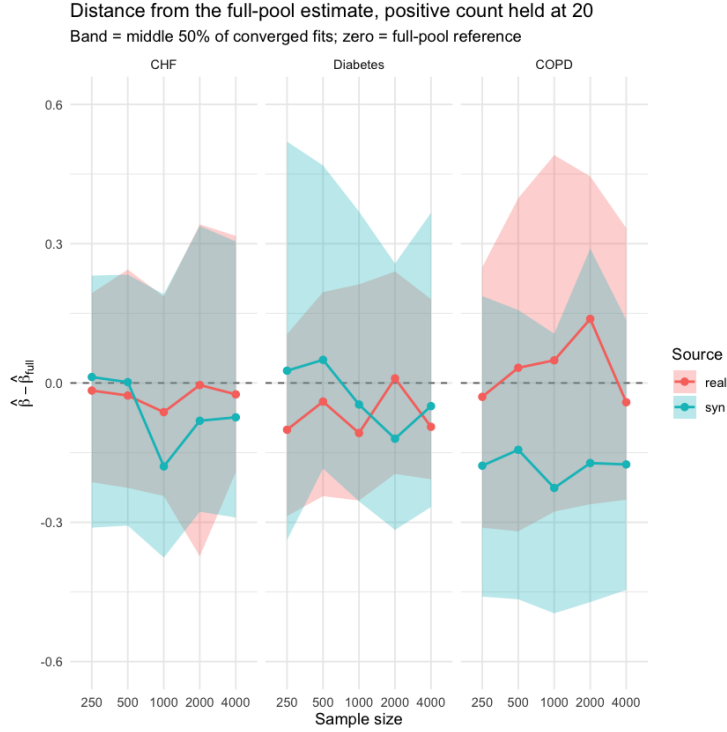


Figure 4: Comorbidity $\times$ creatinine with the positive count held at 20, on the same scale as Figure 2. In the COPD panel the synthetic band sits below zero at every n while the real one straddles it.

Width is not the only thing sample size fails to fix. COPD is the one comorbidity whose interaction is positive, $+0.0668$, and Experiment 1 recovers that sign once the count passes 20. Holding the count at 20 does not: the synthetic median is negative at all five sample sizes, from -0.077 to -0.159 , matching the -0.177 Experiment 1 recorded at the same count. Sixteen times the data leaves the sign where twenty positives put it.

The size effect itself cannot be resolved from noise. The configuration $k = 20$, $n = 500$ appears in both experiments and so is measured twice under independent seeds; the readings differ by 13%, 10% and 0.3%, the same order as the variation across n .

6.3 A Simulation with Known Truth

MIMIC-IV cannot say whether synthesis attenuates the interaction, since there is no truth to compare against and, at these effect sizes, no power to find one. A companion simulation supplies both: a population is generated from a logistic model with a known $\text{chf} \times \text{lactate}$ interaction, and the same count sweep is run at three true values, 200 replicates per cell.

True value	Median estimate		Shortfall	Absolute gap	Median error	
	Real	Synthetic			Real	Synthetic
0.5	0.521	0.496	1%	0.004	0.164	0.219
1.5	1.73	0.891	41%	0.609	0.292	0.713
3.0	2.75	1.22	59%	1.780	0.688	1.90

Table 2: Estimates at $k = 250$ against the value put into the data-generating process. Shortfall and absolute gap compare the median to the truth; the last two columns are the median of $|\hat{\beta} - \text{true}|$ over replicates. Failure at this count has fallen to 0–0.5% in the real arm and 0–4.5% in the synthetic.

The count threshold holds at the two smaller effect sizes, and the synthetic arm clears it later than the

real one at both: divergence reaches zero at $k = 40$ against $k = 80$ at a true value of 0.5, and at $k = 150$ against $k = 250$ at 1.5. At a true value of 3 neither arm reaches zero anywhere on the grid, and the real rate is not even monotone, rising from 10.5% at $k = 5$ to 29% at $k = 20$ before falling to 0.5% at $k = 250$. The reason is the cutoff itself: an estimate pinned near zero cannot exceed it, so at low counts a strong interaction is missed rather than blown up. Convergence there is not success. At $k = 20$ the 71% of real replicates that converge have a median of -0.201 against a truth of 3. What the simulation adds is the size of what survives, and it is not a fixed fraction: the synthetic median recovers 99% of a true value of 0.5, 59% of 1.5 and 41% of 3 (Table 2). The shortfall widens as the target grows and the absolute gap widens faster, from 0.004 to 1.78.

The row at 0.5 is therefore not evidence that small interactions survive synthesis. An estimator that saturates looks accurate against any target small enough, and this one saturates: the three synthetic medians sit close to $0.70\sqrt{\text{true}}$ across the range. The per-replicate errors say the same thing without the fit: at a true value of 0.5 the two medians agree to within 1%, but the median absolute error is 0.219 on the synthetic side against 0.164 on the real, 34% larger. A median that lands on the truth is not the same as estimates that do. Nothing in the table extends to the 0.050 to 0.071 MIMIC-IV offers, an order of magnitude below the smallest value simulated. Recomputing the divergence cutoff at 5, 10, 20 and 40 moves the shortfall by at most five points.

Attenuation is bought, not incurred. At a true value of 3 the real arm diverges in 21–29% of replicates between $k = 20$ and $k = 80$ while the synthetic arm diverges in 2.5–10.5%, the reverse of the pattern on MIMIC-IV everywhere but the two COPD cells of Figure 1. A strong interaction puts the real subsample close to separation; the tree smooths that away, so the synthetic fit converges more readily and converges on the wrong value. At $k = 80$ the real median is 2.85 against a truth of 3, and the synthetic median is -0.150 .

Below the threshold the count governs the sign and not only the spread. The real median carries the wrong sign through $k = 5$ at a true value of 0.5, through $k = 10$ at 1.5 and through $k = 20$ at 3; the synthetic median stays wrong one to two counts longer in each case, through $k = 40$, $k = 40$ and $k = 80$. The larger the interaction, the longer the median points the wrong way, which is the same mechanism read from the other side: a strong effect drives more of the low-count replicates to the boundary, and the ones left converging are the ones that did not find it. It is the pattern the COPD panel of Figure 4 shows on MIMIC-IV, where twenty positives hold the synthetic median at the wrong sign through a sixteenfold increase in n .

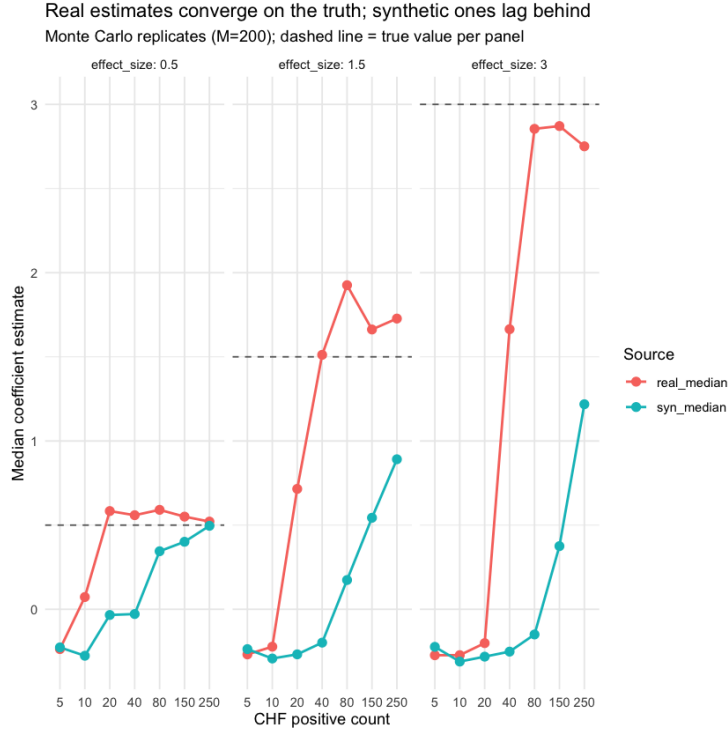


Figure 5: Median estimate against the positive count at three true values, dashed line the truth in each panel. Table 2 reports the $k = 250$ column; the curve shows that the synthetic arm reaches it from below and non-monotonically, passing through -0.150 at $k = 80$ in the rightmost panel.

Taken with the MIMIC-IV sweeps the picture is consistent in one direction and not the other. The count threshold appears wherever it can be looked for, at simulated effect sizes and at clinical ones. Usable recovery appears only at simulated ones.

6.4 Tree Diagnosis: The Diagnostic Needs a Null

Count sweep

What looks like recovery is class balance, not signal. Read without a control the diagnostic rises from 0 at $k = 5$ to 0.26–0.40 at $k = 250$, but the permutation null rises with it, from 0.00 to as high as 0.32 (Table 3). What grows alongside the count is the balance of the column itself, from 1% positive at $k = 5$ to 50% at $k = 250$, and a balanced binary variable is structurally easy to split on whether or not it carries signal [9]. Any diagnostic of this form requires permutation calibration before it can be read. The rates in both tables are proportions over 100 replicates and are less precise than they look: 0.27 carries a 95% interval of $[0.19, 0.37]$, and 0.00 bounds the rate at 0.036 rather than establishing it as nil.

k	CHF		Diabetes		COPD	
	Observed	Null	Observed	Null	Observed	Null
5	0.00	0.00	0.00	0.00	0.01	0.00
10	0.04	0.01	0.00	0.01	0.04	0.01
20	0.08	0.01	0.02	0.04	0.09	0.05
40	0.21	0.07	0.15	0.08	0.16	0.07
80	0.21	0.23	0.27	0.15	0.30	0.19
150	0.28	0.28	0.28	0.32	0.28	0.21
250	0.27	0.30	0.40	0.20	0.26	0.27

Table 3: Proportion of replicates in which CART selects the comorbidity as a split, against its permutation null.

Read across the grid rather than cell by cell, the effect is clear. A conditional logistic model with each replicate as its own stratum puts the real column ahead of the shuffled one by an odds ratio of 2.0 at $k = 40$ ($p = 6 \times 10^{-6}$), and that advantage narrows as the count rises, by a factor of 0.81 per doubling ($p = 0.015$). Fitted across the sweep it runs from 3.7 at $k = 5$ to 1.2 at $k = 250$, which is the null closing on the observed rate rather than the observed rate pulling away. CHF and COPD show both terms on their own; Diabetes shows neither.

The same data supports much less one cell at a time. Most replicates are concordant and carry no information about which column is better, so each paired test turns on a handful of discordant pairs: $p < 0.01$ in two of the 21 cells, one surviving a Bonferroni correction, against 249 discordant replicates to 164 when the grid is pooled.

Size sweep

Holding the count at 20 runs class balance the other way, from 8% positive at $n = 250$ to 0.5% at $n = 4000$, and the null falls with it, ending between 0.00 and 0.03 (Table 4). The null therefore tracks the balance of the column and nothing else, in a range the count sweep never reached. The observed rate does not follow it down, staying between 0.04 and 0.14 at every size, where the count sweep left it at $k = 20$.

n	Prevalence	CHF		Diabetes		COPD	
		Observed	Null	Observed	Null	Observed	Null
250	8.0%	0.05	0.03	0.07	0.02	0.06	0.08
500	4.0%	0.04	0.02	0.08	0.04	0.06	0.01
1000	2.0%	0.14	0.02	0.07	0.02	0.08	0.04
2000	1.0%	0.06	0.01	0.12	0.02	0.06	0.02
4000	0.5%	0.07	0.00	0.09	0.01	0.04	0.03

Table 4: CART’s use of the comorbidity against its permutation null, with the positive count held at 20.

With the null near zero the signal becomes legible. Pooled over this grid the real column wins 104 discordant replicates to 32, and 14 of the 15 cells point the same way, against 249 to 164 in the count sweep. Per cell the tests stay underpowered, only CHF at $n = 1000$ surviving a Bonferroni correction over 15 comparisons, but the direction is no longer in doubt: the diagnostic is readable where its null is near zero, which on this data means low prevalence.

Neither sweep implicates CART on its own. A GLM fitted to the same subsamples detects the comorbidity in 2–16% of replicates in every cell of both sweeps, against the 6–7% predicted by rescaling the full-pool standard errors to $n = 500$, and it does no better on real labels than on shuffled ones: over the size sweep

the observed rate averages 9.1% and the permuted 9.5%, with no trend in n . Both methods are blind at these sizes; CART is the one that fails visibly, by declining to spend a split.

7 Discussion

Two failure modes, not one. Reporting the failure rates separately from the spread changed the reading of this data more than any other decision. Mixing them inflates apparent instability at low counts, since divergent fits push the quantiles outward, and it hides the clearest result in the study, a failure threshold that every series reaches between $k = 40$ and $k = 80$.

A privacy-fidelity tension. CART’s reluctance to isolate a subgroup with few supporting cases is not simply a limitation to work around. A tree willing to carve out a node of five patients would also be more likely to overfit to, and risk re-identifying, those individuals, and that same reluctance is what leaves the stratum missing from the synthetic table in the inestimable cases at $k = 5$. The tension is a property of the method rather than an implementation defect, though only its fidelity side is measured here.

The design, not the generator. A 500-row subsample carries 6–7% power against these interactions and the GLM arm returned 6.7–10.1%, both within a few points of the 5% a test with no power at all would produce. Theory and experiment landing in the same place is what rules out a defect in the pipeline and locates the failure in the design rather than in the generator. The size of the gap is easier to read as a detectable effect than as a rate: at $n = 500$ the smallest interaction detectable at 80% power is 0.472 for CHF, 0.485 for Diabetes and 0.775 for COPD, against the 0.0497, 0.0709 and 0.0668 actually present, a factor of 9.5, 6.8 and 11.6. Reaching those interactions instead of that floor would need between 23,384 and 67,238 rows. The floor also explains a choice in Section 6.3 that would otherwise look arbitrary: the smallest interaction put into the simulated data-generating process, 0.5, is of the same order as the smallest this design can detect at all, so the simulation starts where MIMIC-IV’s detectable range ends. The two data-generating processes differ, so the floors are comparable in scale and not in value.

Limitations. The diagnostic tree is a proxy for the generator: three predictors rather than the full feature set `synthpop` synthesizes jointly, matched on control parameters but not on its sequential visit order, and counting the comorbidity as used wherever it appears rather than requiring it beneath a creatinine split. All three are permissive, which makes non-selection the more reliable of the two outcomes. Monte Carlo error at $M = 100$ reaches 13% on interquartile widths, and the paired tree tests turn on at most 40 discordant replicates per cell in the count sweep and 14 in the size sweep, so a cell that misses significance is not evidence that nothing is there. Divergence at the smallest counts is largely separation: a penalized fit removes 60–70% of it at $k = 5$ and most of the rest by $k = 20$, but it does not move the count at which divergence stops, which stays at 40 for CHF and COPD under both fits. The threshold is not an artifact of the estimator, though what sets its location is not established here. One continuous variable and one outcome were examined, Renal Failure and Hypertension were screened but not analyzed, the pool excludes the short, high-mortality stays that carry no first-day creatinine, and all results come from a single MIMIC-IV snapshot.

8 Conclusion

The positive count governs every quantity measured here, and sample size governs none of them. Divergence falls to zero between counts of forty and eighty in all three comorbidities and both data sources; interquartile width falls by 42–66% over a matched increase in the count; CART’s use of the comorbidity is flat across a sixteen-fold change in n ; and a logistic fit to the same subsamples shows no consistent separation from its permutation null under either lever. Four readings on the same replicates, so not four independent

confirmations, but two of them come from data the synthesizer never touched. In that narrow sense the finding from Bayesian logistic regression does extend to CART-based synthesis.

The wider claim does not survive contact with clinical effect sizes. The simulation shows what recovery looks like when the target is large enough to reach, and it is partial: at a true value of 3 the synthetic median reaches 1.22. At practical subsample sizes neither arm recovers the MIMIC-IV interactions at all: even the narrowest cell spans nearly four times the quantity being estimated, and with the count held at 20 the COPD synthetic median carries the wrong sign at every sample size tested. Where the two can be compared, the synthetic side is worse on both failure and spread.

For synthetic data evaluation, two things follow: establish that the target effect is estimable in the source data before attributing a discrepancy to the generator, and calibrate any variable-selection diagnostic against a permutation null, since its null moves with class balance from zero to 0.32 across the range tested here.

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